

COURSE NAME & CODE: ECA 0303 – Signals and systems

NAME: D S Lakshmi Narayana

REG NO: 192512371

Experiment 4.2

Causal/Non-Causal System

Aim:

To analyze and distinguish between causal and non-causal discrete-time systems. The Aim is to understand how causality affects the system's response to different input sequences.

Program:

% Test Case: Causal and Non-Causal Systems Analysis

clc;

clear;

close all;

% Define parameters

N = 20; % Number of samples

n = 0:N-1; % Discrete time index vector

% Input sequence: Unit step sequence

x_n = ones(1, N);

% Initialize output sequences

y_causal = zeros(1, N);

```
y_noncausal = zeros(1, N);
```

```
%% Causal System:  $y[n] = x[n] + 0.5x[n-1]$ 
```

```
for k = 1:N
```

```
    if k > 1
```

```
        y_causal(k) = x_n(k) + 0.5 * x_n(k-1);
```

```
    else
```

```
        y_causal(k) = x_n(k); % No  $x[n-1]$  for first sample
```

```
    end
```

```
end
```

```
%% Non-Causal System:  $y[n] = 0.5x[n] + 0.5x[n+1]$ 
```

```
for k = 1:N
```

```
    if k < N
```

```
        y_noncausal(k) = 0.5 * x_n(k) + 0.5 * x_n(k+1);
```

```
    else
```

```
        y_noncausal(k) = 0.5 * x_n(k); % No  $x[n+1]$  for last sample
```

```
    end
```

```
end
```

```
%% Plot the input and output sequences
```

```
figure;
```

```
% Input sequence
```

```
subplot(3,1,1);
```

```
stem(n, x_n, 'filled');
```

```
grid on;
```

```
xlabel('n (Discrete Time Index)');
```

```
ylabel('Amplitude');
```

```
title('Input Sequence: Unit Step Sequence');
```

```
% Output of the causal system
```

```
subplot(3,1,2);
```

```
stem(n, y_causal, 'filled');
```

```
grid on;
```

```
xlabel('n (Discrete Time Index)');
```

```
ylabel('Amplitude');
```

```
title('Output Sequence: Causal System');
```

```
% Output of the non-causal system
```

```
subplot(3,1,3);
```

```
stem(n, y_noncausal, 'filled');
```

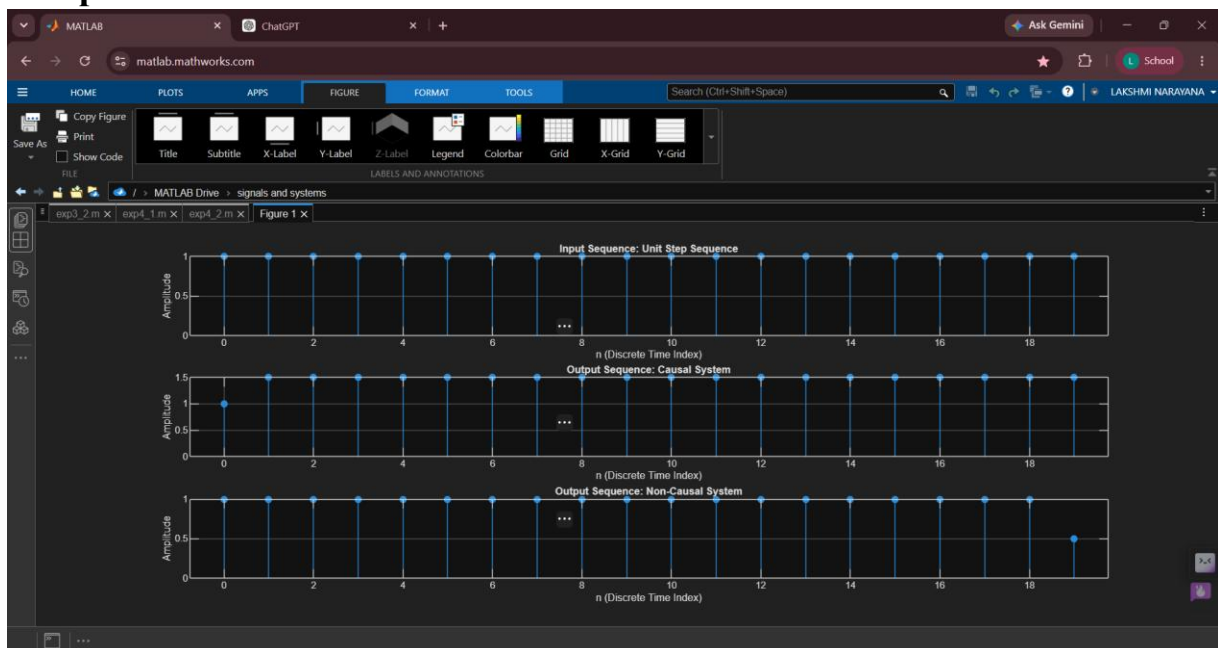
```
grid on;
```

```
xlabel('n (Discrete Time Index)');
```

```
ylabel('Amplitude');
```

```
title('Output Sequence: Non-Causal System');
```

Output:



Result:

- Causal System: The causal system only depends on the current and past input values. The output does not require future input values, making it physically realizable in real-time systems.
- Non-Causal System: The non-causal system depends on future input values, which is not physically realizable in real-time systems but useful for certain theoretical analyses.

This test case helps to illustrate the difference between causal and non-causal systems, demonstrating how their characteristics affect the response to a given input sequence